Experiment (1)
Semiconductor Material Bandgap

Objectives

- To study the relationship between a material bandgap and critical absorption wavelength.
- To identify the color emitted by five different LEDs using various light sources.

Pre-lab Work

Review the photon absorption in semiconductors and the critical wavelength equation.

Circuit

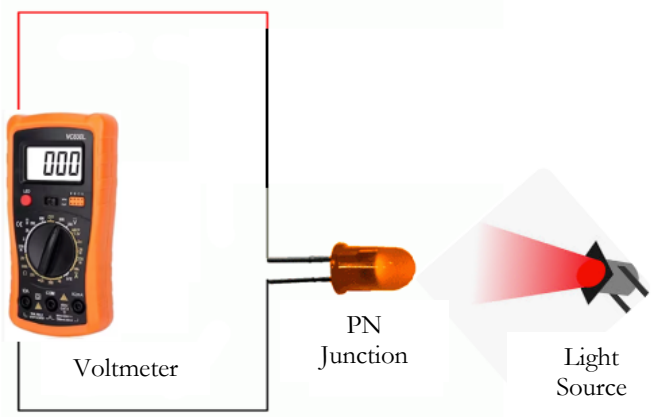


Figure 1: Circuit for bandgap identification.

Equipment

Table 1. Equipment for experiment 1.

	Equipment	Model	Specifications
1	BreadBoard Station	--	
2	Light sources (3)	--	Red, Green and White.
3	Assorted PN junctions (5)	--	1-4 Volts, Up to 30 mA (labeled 1-5)
4	Multi-meter (1)	--	
5	Cables and jumpers	--	

Procedure

1. Connect the circuit in figure 1. For each PN junction, illuminate with the available light sources and measure the potential difference developed across the PN junction. Make sure to get the maximum response by slightly adjusting the light source spot of the PN junction. Record your results in table 1.
2. Repeat for all available PN junctions, each illuminated by the three light sources.

Results and Analysis

1. Study the data in the first three row of table 1 in connection with the critical absorption wavelength and its relation to the band gap.

Table 1. Light-induced potential across PN junctions.

	Maximum Potential Difference (V)				
Light Source	PN junction 1	PN junction 2	PN junction 3	PN junction 4	PN junction 5
IR	107mV	0mV	0V	1680mV	0V
Green	1645mV	0mV	0V	1560mV	1840mV
White	1671mV	2320mV	0V	1558mV	2010mV
PN Junction Color	Green	White	UV	White	Green
PN Junction Material	InGaN	InGaN+ AlInGaP	GaN	InGaN+ AlInGaP	InGaN

2. In the last row of table 1, enter the most probable color of each PN junction. Give your reasons and calculations below.

At the first PN Junction, Color of the PN Junction Material is the most probably green. Because, the highest level of voltage exist in this light spectrum.And also white light include the green light inside it. So that the voltage level is very close to green.It's a little bit higher because of there are also external light exist.

At the second PN Junction,Obviously seen that this is white ligth.Because there is no effect of the green and red voltage for generating voltage on multimeter.However when white light falling on the led, voltage difference occured.

At the third PN Junction , There is no reflection all type of the our light. That's mean the most probably this Photodiode not used for Visible Light Range. Maybe this is usable for Ultraviola Light Spectrum.

At the fourth PN Junction, As seen clearly the voltage value of the Red and Green is very close to each other. So , we can say that White PN Junction Color exist here, because White Light included both of them.

At the last PN Junction , Clearly we say that this is green junction.However, Green intensitiy of the White light is higher than Normal Green light intensity, so voltage difference a little bit higher.